AES 423_ Drainage Technology

General Objectives:

1 Understand agricultural drainage problems and drainage systems terminologies

1.1 Drainage problems

1.1.1 Describe land drainage problems in Nigeria

1.1.2 Define drainage system terminologies e.g. seepage, tiles, leaching, trenches, manholes, channels etc

1.1.3 Explain the importance of drainage in agricultural soils.

2 Understand principle of drainage design

2.1 Drainage principle

2.1.1 Explain the principles of surface and sub-surface drainage

2.1.2 Explain the principles of buried of buried drain and open ditch.

2.1.3 Explain the various elements of sub-surface drainage.

3 Understand field and soil designs for drainage designs

3.1 Soil physical characteristics

3.1.1 Discuss types of investigations surveys (reconnaissance, preliminary and design surveys) for drainage.

3.1.2 Compute soil permeability K

3.1.3 Calculate hydraulic conductivity.

3.1.4 Relate soil conductivity to soil properties.

4 Understand surface drainage systems

4.1 Identify and describe the various types of surface drains.

4.1.1 Explain the function of surface drains.

4.1.2 Describe the construction methods of surface drains.

4.1.3 Explain the importance of land forming in surface drainage.

4.1.4 Explain the need for joint surface and sub-surface drainage systems.

4.1.5 Maintain surface drainage systems.

5 Understand sub-surface drainage systems

5.1 Sub-surface drainage

5.1.1 Explain the procedure for carrying out sub-surface drainage

5.1.2 Install and maintain sub-surface drains

DEPARTMENT OF AGRICULTURAL AND BIOENVIRONMENTAL ENGINEERING TECHNOLOGY
(FEDERAL POLYTECHNIC, MUBI)

COURSE TITLE:	**Drainage Technology (AES 423/AES 432)**
COURSE LECTURER(S):	Dr Mohammed Umar Abba Dr Emmanuel Kefas Bwade

1 DRAINAGE & DRAINAGE PROBLEMS (Lecture No. 1)

Drainage is the process of removal of excess water as free or gravitational water from the surface of farm lands with a view to avoid water logging, creates favourable soil conditions for optimum plant growth, and enhance agricultural production of crops. It may involve any combination of storm water control, erosion control and water table control.

Drainage is a general belief by the public that drainage is not needed in arid regions of the world especially that waterlogging is not common in arid regions. Yet the truth of it is that drainage is required in arid regions especially where there is over irrigation or seepage for riverbanks or lakes. This is necessary to maintain optimum soil water and salt balance that is required for optimum crop performance. As a matter of facts, drainage is required under the following conditions –

1. In areas of high water table
2. Water ponding under on soil surfaces for days, weeks or months periods
3. Excessive soil moisture content above the field capacity on heavy soils such as clay soils.
4. Areas of salinity and alkalinity where annual evaporation exceeds rainfall and capillary rise of ground water occurs.
5. Humid regions with continuous or intermittent heave rainfall.
6. Flat land with fine textured soils.
7. Low-lying flat areas surrounded by hills.

1.1 Land Drainage Problems in Nigeria

Drainage problems in Nigeria pose a significant challenge to agricultural productivity, particularly in regions with poorly drained soils, high rainfall, and improper land use. One of the most common problems is ***waterlogging***, which occurs when soil becomes saturated with water, displacing air in the root zone. This condition prevents adequate oxygen supply to plant roots, leading to reduced crop yield, root rot, and poor plant development. Waterlogging is frequently observed in flat lowland areas such as inland valley bottoms, floodplains, and poorly drained clayey soils, especially in parts of the Niger Delta and Middle Belt.

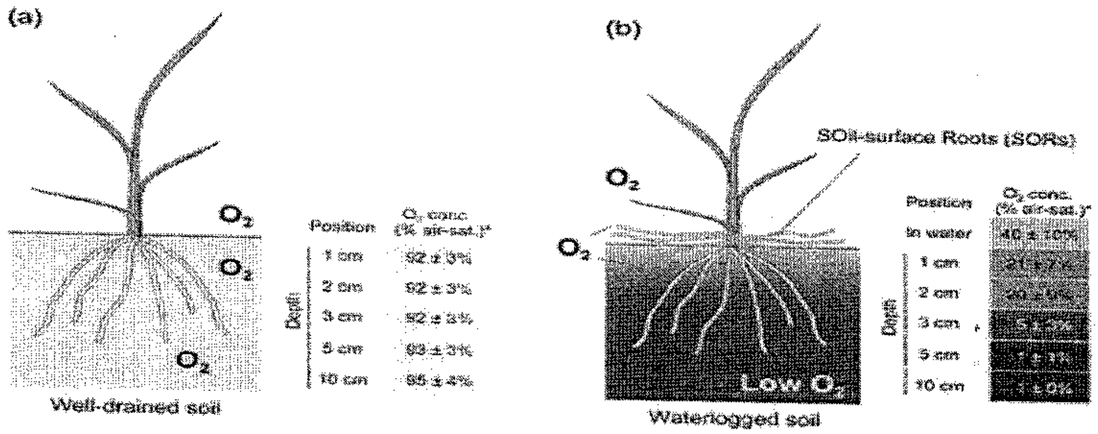

Figure 1. Sketch of well-drained and waterlogged soils

From Figure 1, it can be seen that the entire root zone of rice crop (10 cm) contains considerable concentration of oxygen (up to 95% at the depth of 10cm). While waterlogged soil has limited concentration of oxygen even near the soil surface (40%) with only 1% of oxygen concentration observed at the depth of 10 cm from the soil surface.

Another serious issue is ***flooding***, which affects both urban and rural agricultural lands in Nigeria. Flooding is typically caused by intense and prolonged rainfall, overflowing rivers, inadequate or blocked drainage systems, and uncontrolled land development. For instance, many agricultural communities in Lagos, Anambra, and parts of Adamawa State experience seasonal flooding that destroys crops and farmland infrastructure. Inadequate drainage planning exacerbates this problem. Examine the soil drainage map in Figure 2.

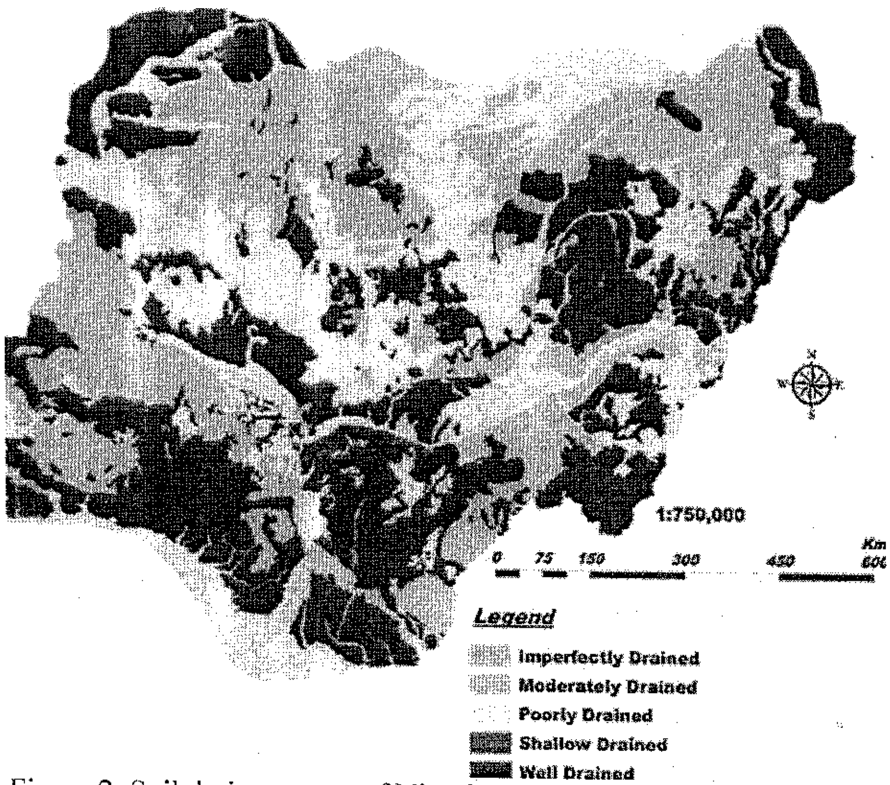

Figure 2. Soil drainage map of Nigeria (Source: Soil Digital Map/GIS)

The presence of a ***high water table*** is also a concern, especially in coastal areas and swampy regions. When the water table rises too close to the surface, it limits root growth and makes soil too soft for mechanized farming. Furthermore, in the semi-arid northern regions such as Yobe and Borno States, poor drainage systems and excessive irrigation without proper leaching lead to *soil salinization*. This occurs when dissolved salts accumulate in the root zone due to evaporation, rendering the soil unproductive over time. Similarly, *seepage problems* are common around irrigation canals and earthen reservoirs, where water slowly infiltrates surrounding soils, resulting in saturation and in some cases, erosion.

An increasingly severe consequence of poor surface drainage is ***gully erosion***. This is especially evident in the South-Eastern states like Anambra and Imo, where uncontrolled runoff and absence of check structures lead to the formation of deep gullies that damage farmlands and settlements. Overall, these drainage problems reduce cultivable land, lower crop productivity, damage infrastructure, and undermine environmental sustainability.

1.1.1 Drainage System Terminologies

In order to understand and effectively manage drainage systems, it is important to become familiar with key terminologies commonly used in drainage engineering such as seepage, tile drains, leaching, trenches, manhole, channels, water table, and drainage coefficient.

i. ***Seepage*** refers to the slow movement of water through soil pores, either downward or laterally. It is especially important in areas where irrigation canals or water bodies are adjacent to cultivated land, as seepage can lead to waterlogging or instability of slopes.

ii. ***Tile drains***, also known as ***subsurface drains***, are perforated or porous pipes laid underground to collect and transport excess water away from the root zone. Traditionally made from clay, modern tile drains are usually constructed from PVC or polyethylene.

iii. ***Leaching*** is the process by which water percolates through the soil, dissolving and carrying away soluble salts and nutrients beyond the root zone. While leaching helps in salinity control, it may also lead to nutrient loss if not properly managed.

iv. ***Trenches*** are excavated channels or ditches, typically dug to lay pipes or guide surface runoff. In subsurface drainage, trenches are backfilled with permeable materials to aid water infiltration.

v. ***Manholes*** are vertical inspection chambers built into drainage networks to allow for monitoring, cleaning, and maintenance of underground drainage lines. These are essential in large-scale drainage installations.

vi. ***Channels*** are open conveyance structures—natural or artificial—used to direct surface water away from farm fields to safe outlets. These include ditches, grassed waterways, and lined canals.

vii. ***Water table*** refers to the upper boundary of the saturated zone in the soil. In well-drained land, the water table lies below the root zone, while in poorly drained land, it may rise close to the surface.

viii. ***Percolation*** is the term that describes the vertical movement of water through soil layers and plays a critical role in both infiltration and groundwater recharge.

ix. ***Drainage Coefficient (DC)*** is a design parameter that expresses the depth of water (in mm) that must be removed from a field per day to prevent waterlogging. It is used to determine the size and capacity of surface and subsurface drains.

1.2 Importance of Drainage in Agricultural Soils

The importance of drainage in agricultural soils cannot be overstated, especially in a country like Nigeria where climate conditions, soil types, and topography vary widely. One of the primary benefits of effective drainage is improved ***root aeration***. Healthy root systems require oxygen for respiration and nutrient uptake. In waterlogged conditions, air is displaced by water, leading to oxygen deficiency, root decay, and stunted growth. Proper drainage lowers the water table and improves oxygen availability in the root zone.

Drainage also prevents ***waterlogging***, a major constraint to crop production in flat and poorly structured soils. By removing excess water from the surface and subsoil, drainage systems help maintain an optimal moisture balance for crop growth. Well-drained soils are more *workable*, meaning that they can be tilled, planted, and harvested with greater ease, especially with the use of mechanized equipment. In contrast, waterlogged soils become too soft or sticky, increasing fuel consumption and wear on machinery.

Another significant function of drainage is the *leaching of salts* from the root zone. This is particularly critical in irrigated fields in arid and semi-arid zones, where evaporation can cause salt build-up. With adequate drainage, salts are flushed below the root zone, preventing salinity-related damage to crops. In addition, well-drained soils tend to *warm up faster* in the early part of the season, leading to improved germination and early crop development.

Drainage also contributes to *erosion control*. Surface runoff that is not properly managed can cause sheet, rill, or gully erosion. Well-designed surface drainage systems help reduce the volume and velocity of runoff, thereby minimizing soil loss and preserving land quality. Furthermore, improved drainage reduces the *incidence of plant diseases* and pests. Stagnant water provides a breeding ground for fungal pathogens, nematodes, and mosquitoes, all of which negatively affect crops and human health.

Perhaps most importantly, proper drainage *increases crop yield and quality*. Studies have shown that well-drained fields can yield 30–50% more than poorly drained fields, especially for water-sensitive crops like maize, cowpea, and groundnuts. In rice production, controlled drainage is essential for managing paddy fields. In summary, efficient drainage is fundamental to sustainable agricultural development, soil conservation, and food security.

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2 CHARACTERISTICS & TYPES OF DRAINAGE SYSTEMS (Lecture No. 2)

2.1 Characteristics of good drainage system

A good drainage system should consist of the following characteristics –

1. It must be permanent.
2. It must have adequate capacity to drain the area completely.
3. There should be minimum interference with cultural operations.
4. There should be minimum loss of cultivable area.
5. It should intercept or collect water and remove it quickly within shorter period.

2.2 Classification of drainage systems based on soil profile

Agricultural drainage systems can be classified into categories depending on the criterion used for the classification. Based on the depth of water to be removed, drainage can be classified into two –

1. Surface drainage system
2. Sub-surface drainage system

2.2.1 Surface Drainage

The surface drainage is designed primarily to remove excess water from the surface of soil profile. This can be done by developing slope in the land so that excess water drains by gravity; generally, it can be made by smoothening of soil surface and construction of ditches. It is suitable for the following conditions –

1. Slowly permeable clay and shallow soils
2. Regions of high intensity rainfall
3. Fields where adequate outlets are not available.
4. Land with less than 1.5% slope

2.2.1.1 Types of surface drainage systems

The surface drainage system can be classified into four (4) sub-groups/classes, namely, i) lift drainage ii) gravity drainage iii) field-surface drainage and iv) ditch drainage, as shown in the following figure –

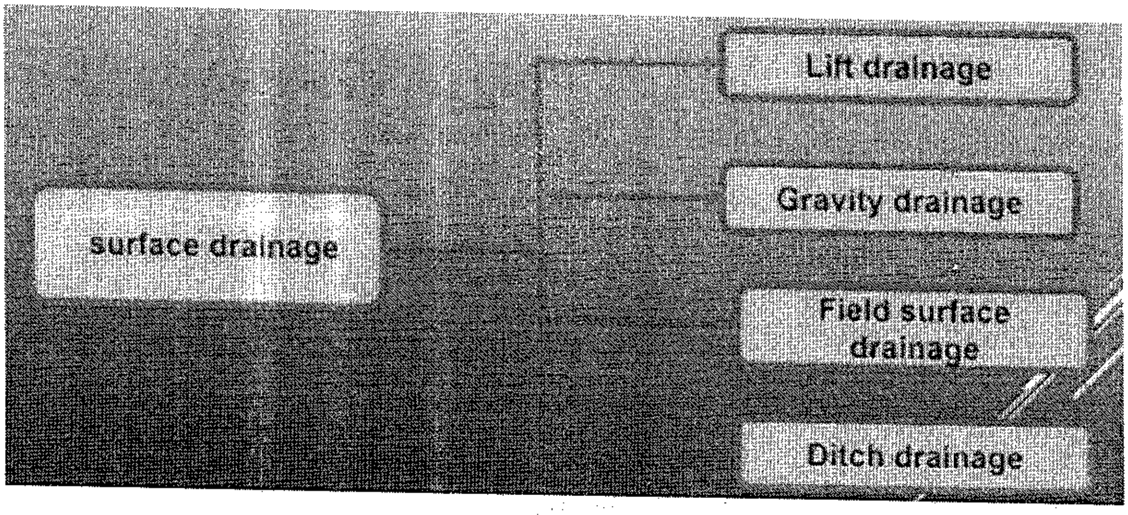

Figure 3. Categories of surface drainage systems

2.2.1.1.1 Lift drainage –

Lift Drainage involves the removal of surface or excess water using mechanical means, such as pumps. It is typically used in low-lying or flat areas where gravity flow is not possible due to embankment. Lift drainage systems require significant infrastructure, including pumps, power sources, and drainage channels or pipes. While it is highly effective and can operate even without natural slope, the cost of installation and maintenance is high due to mechanical components and energy needs. This type of drainage system is suitable for areas with high water tables or where other drainage systems are impractical.

Figure 4. Lift drainage

2.2.1.1.2 Gravity drainage

Gravity Drainage relies on natural land slopes to remove excess water from fields. Water flows due to gravitational pull without the need for mechanical intervention. This method is simple and cost-effective but requires sufficient natural or engineered slope for effective functioning. It works best in undulating or gently sloped terrains. The maintenance requirements are generally low, and when designed properly, gravity drainage can be efficient and long lasting. However, it is ineffective in areas that are flat or with considerable depression; where water cannot flow freely.

Figure 5. Gravity drainage

2.2.1.1.3 Field surface drainage

Field Surface Drainage involves reshaping or levelling the land to promote the natural flow of water off the field surface. Techniques such as land grading, constructing shallow furrows, or smoothing ridges are used to direct water away from crop root zones. This type of drainage is moderately efficient and helps reduce waterlogging and soil erosion when slopes are gentle and uniform. Field surface drainage systems require moderate cost and maintenance, as periodic reshaping or regrading may be necessary. It is particularly useful in areas where rainfall causes temporary ponding.

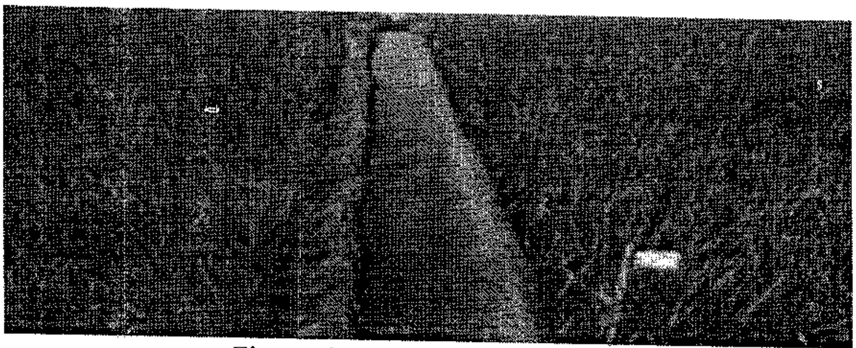

Figure 6. Field surface drainage

2.2.1.1.4 Ditch drainage

Ditch Drainage uses shallow open channels (ditches) constructed across or along the field to carry away excess surface water. These ditches intercept runoff and guide it toward a main outlet or waterway. While ditch drainage has a low initial cost and is simple to implement, its efficiency is generally low. Water may take time to reach the ditch, especially on flat terrain, and ditches are prone to clogging by weeds, sediment, or debris. They also reduce usable farmland and require frequent maintenance. Despite these limitations, ditch drainage is commonly used in traditional agriculture and areas with moderate rainfall.

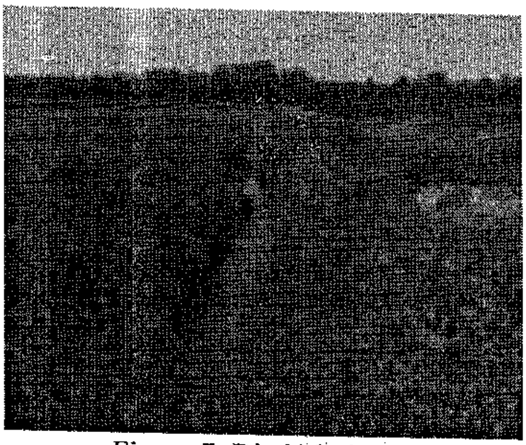

Figure 7. Ditch drainage

2.2.2 Sub-surface drainage system

Sub-surface drainage drains are underground artificial channels through which excess water may flow to a suitable outlet. The purpose is to lower the ground water level below the root zone of the crop. The movement of water into subsurface is influenced by the hydraulic conductivity of the soil, depth of drain below ground surface and the horizontal distance between individual drains. Underground drainage is mostly needed for medium textured soils, high value crops and high soil productivity.

The **sub-surface drainage** is associated with the following **advantages** –

1. There is no loss of cultivable land
2. No interference for field operation
3. Maintenance cost is less
4. Effectively drains sub-surface soil and creates better soil environments.

However, it is associated with the following **disadvantages** –

1. Very high initial cost
2. It requires constant attention
3. It is effective for soils having low permeability.

2.2.2.1 Classification of sub-surface drainage system

There are three (3) categories of the sub-surface drainage system, namely, i) tile drainage ii) mole drainage iii) vertical drainage/drainage wells

2.2.2.1.1 Tile drainage

This consists of continuous line of tiles laid at a specific depth, grade so that the excess water enters through the tiles, and flow out by gravity. Laterals collect water from soil and drains into sub-main and then to main and finally to the outlet. The drains are made with clay and concrete. The tile should be strong enough to withstand the pressure and resistant to erosive action of chemicals in soil water.

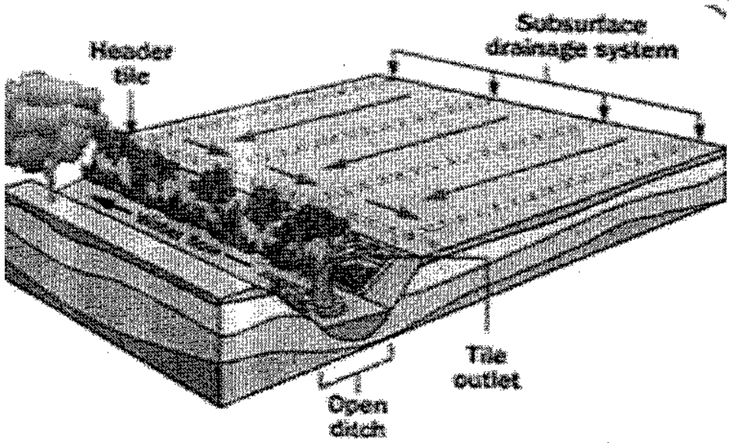

Figure 8. Tile drainage system

2.2.2.1.2 Mole drainage

Mole drains are unlined circular earthen channels formed within the soil by a mole plough. The mole plough has a long blade-like shank to which a cylindrical bullet-nosed plug is attached, known as mole. As the plough is drawn through the soil, the mole forms the cavity to a set depth. Mole drainage is not effective in loose soil since the channels produced by the mole will collapse. This is also not suitable for heavy plastic soil where mole seals the soil to the movement of water.

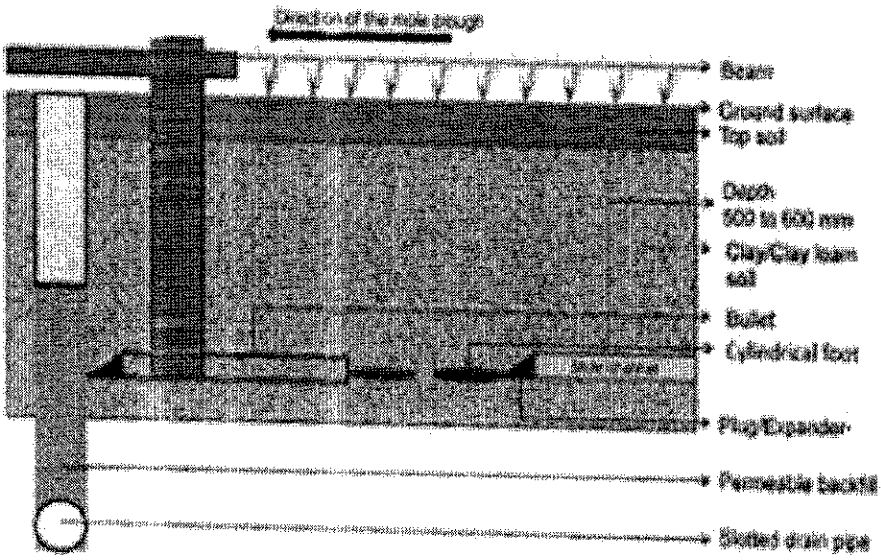

Figure 9. Mole drain system

2.2.2.1.3 Vertical drainage/drainage well

Vertical drainage is the disposal of drainage water through well into porous layers of earth. Such a layer must be capable of taking large volume of water rapidly. Such layers are in riverbeds.

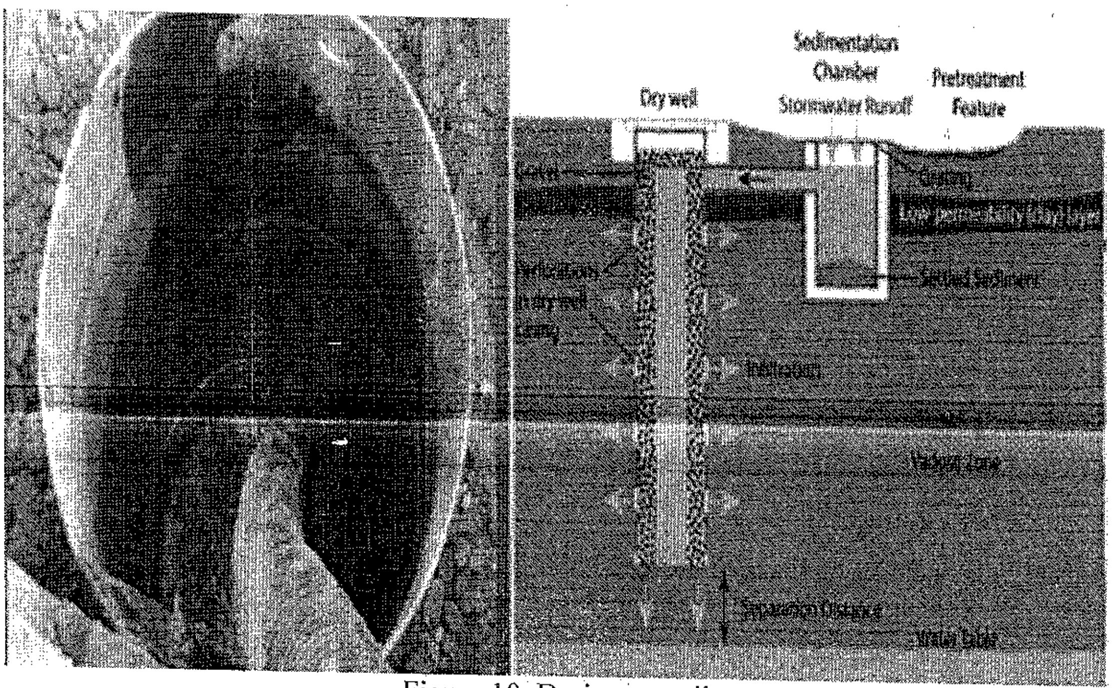

Figure 10. Drainage well

2.3 Classification of drainage system based on layout

Regardless of whether the drainage system is to be implemented on the soil surface or sub-surface, it can be classified based on the layout the structure assumes during or after installation. Moreover, based on this, it can be classified into four (4) groups/classes, namely, i) Random ii) Herringbone, iii) Gridiron and iv) Interceptor systems.

2.3.1 Random drainage system

Random Drain System is a flexible drainage layout used primarily in fields where wet spots are scattered irregularly, such as in undulating or uneven terrain. In this system, drains are installed only where needed—targeting isolated waterlogged areas—without following a strict geometric pattern. This approach minimizes unnecessary installation and cost, making it practical in situations where drainage problems are not uniformly distributed. However, construction requires careful planning, as the layout depends entirely on the topography and specific locations of wetness. Maintenance can be challenging due to the irregular arrangement of drain lines, which may not be easily accessible or follow a predictable route.

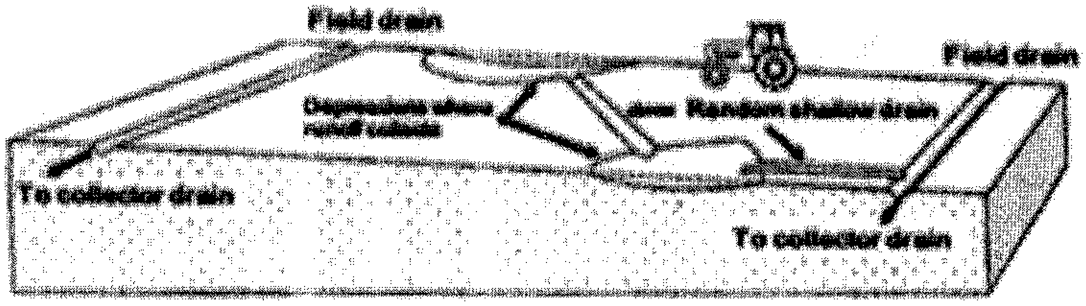

Figure 11. Random drain system

2.3.2 Herringbone system

Lateral drains entering the main drain from both sides at an angle characterize herringbone System, creating a layout that resembles a fish skeleton or herringbone. This system is best suited to fields with a uniform slope towards a central main drain or a shallow depression that collects water. It ensures efficient water collection from both sides of the main drain, especially in moderately sloping areas. The herringbone design offers better coverage than random systems and is easier to manage due to its semi-structured nature. However, it requires precise planning and moderate construction effort, particularly in ensuring symmetrical alignment of laterals. Maintenance is manageable because of the organized and angled drain pattern.

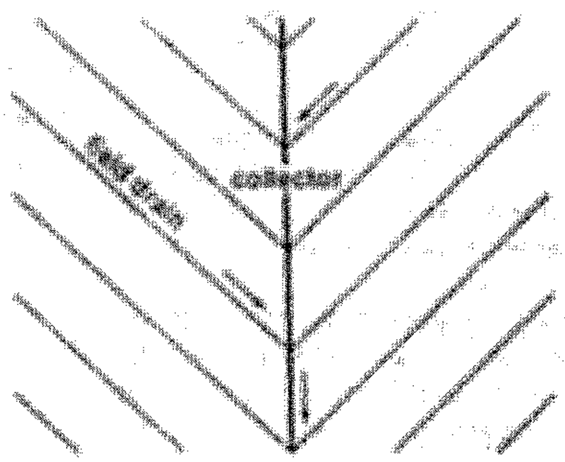

Figure 12. Herringbone system

2.3.3 Gridiron system

This system is also known as the parallel system, features lateral drains running at right angles into a straight main drain, forming a rectangular grid-like pattern. This layout is ideal for flat or uniformly sloped fields and is commonly used in areas with widespread drainage needs. The gridiron system is simple and efficient, allowing uniform drainage across the entire field. It is easy to construct and maintain due to its repetitive and organized structure, and it generally has a lower cost compared to systems that are more complex. The straight, evenly spaced lines allow for easy access for inspection and repairs. Overall, the gridiron system is highly effective in large, level agricultural fields requiring comprehensive drainage.

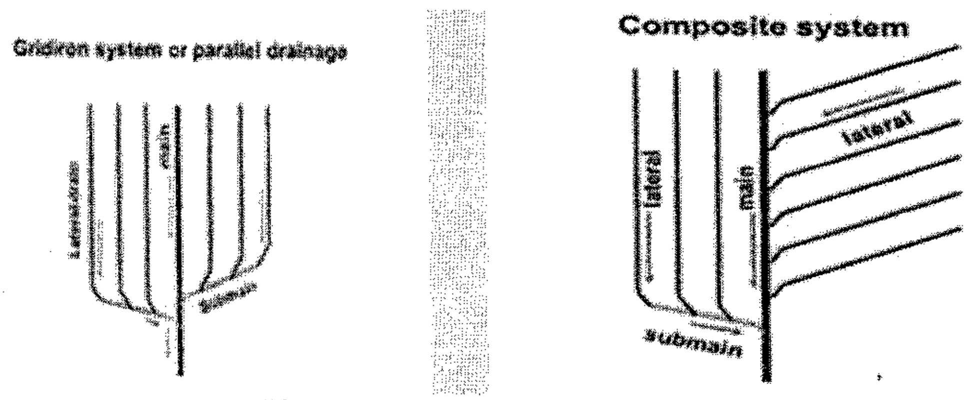

Figure 13. Gridiron drainage system

2.3.4 Interceptor

Interceptor ditches are a specialized type of surface drainage designed to intercept and divert surface runoff before it enters or accumulates in a particular area, such as a field, road, construction site, or low-lying land. These ditches are typically constructed along the upper boundaries or slopes of an area to capture runoff from higher ground and redirect it safely to a suitable outlet, thereby preventing erosion, flooding, or waterlogging in the protected area.

The layout of interceptor ditches is often linear or slightly curved, following the natural contour lines of the terrain to minimize water velocity and reduce erosion. They are especially useful in hilly or sloped areas where uncontrolled runoff can cause significant soil degradation or interfere with agricultural operations. Interceptor ditches do not drain the soil within the field itself but protect the area by managing external water sources.

Construction of interceptor ditches requires careful surveying to ensure they are placed at the right elevation and gradient to collect and convey water effectively without overflowing. While the cost of construction is generally moderate, they need periodic maintenance, such as clearing of silt, vegetation, and repairing erosion damage. Despite being a simple and cost-effective solution, their effectiveness depends on proper design and regular upkeep. In agricultural settings, interceptor ditches play a preventive role, complementing internal drainage systems like random or gridiron layouts.

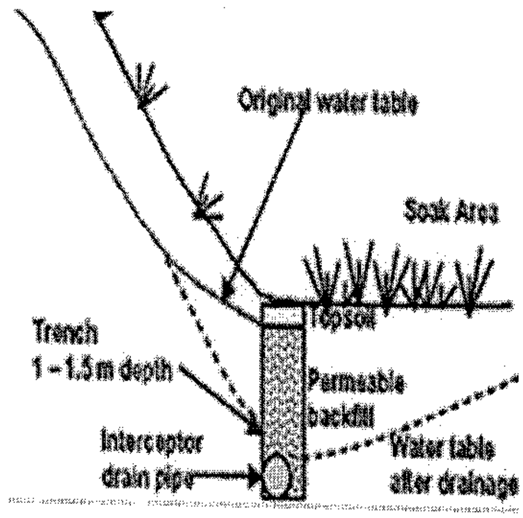

Figure 14 Interceptor drain

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3 Drainage design Variables (LECTURE NO. 3)

3.1 Soil physical characteristics

3.1.1 Discuss types of investigations surveys (reconnaissance, preliminary and design surveys) for drainage.

3.1.2 Compute soil permeability K

3.1.3 Calculate hydraulic conductivity.

3.1.4 Relate soil conductivity to soil properties.